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Compost Amendments and Water Infiltration in Compacted Urban Soils

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Abstract. Compacted urban soils absorb little of the rain that falls on them. We tested whether a single compost amendment changes how much simulated rainfall a plot takes in.

Methods

Eighteen plots on a former car park were loosened by hand; half received a 5 cm layer of compost worked into the top layer. A rainfall simulator delivered a steady shower to each plot, and run-off was collected and measured.

Results

Averaged across the nine treated plots and measured from the moment the shower began, compacted plots with compost amendments absorbed 31 mm of simulated rainfall in the first hour, against 9 mm on untreated plots. The difference persisted when the test was repeated the following spring.

Discussion

Compost improves the structure of the soil rather than its surface, so its effect lasts beyond the season it is applied. It pairs well with a mulch, which keeps the absorbed water from evaporating.